Technical Data Sheet

(TDS)

TINMORRY Metallic PETG

Version: V1.0

• Introduction

TINMORRY Metallic PETG is a PETG filament designed to create a distinctive metallic appearance. Both the filament and printed parts have a smooth, refined surface with a subtle metallic sheen. As the viewing angle changes, variations in light and shadow across the surface create a realistic metal-like visual effect.
It also retains the excellent toughness and interlayer bonding strength of standard PETG, together with good water resistance, oil and grease resistance, and weather resistance. It supports relatively high printing speeds and is less prone to nozzle buildup, oozing, stringing, or bubbling during printing. It is compatible with most FDM 3D printers and is available in a wide range of colors, offering excellent value. TINMORRY Metallic PETG is therefore a creative filament that delivers a realistic metallic appearance at a lower cost while combining practical performance with decorative appeal.

• Product Specifications

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Weight	1000 g
Length	320 ± 5 m
Spool Material	PP plastic (heat resistant to approx. 100 °C)

Spool Dimensions	Outer diameter: 200 mm; hub opening diameter: 55 mm; Width: 62 mm

• Recommended Printing Parameters

Pre-print Drying Conditions	65-70 °C, 6-8 h (forced-air oven or filament dryer)
Printing Environment	≤ 60 °C, ≤ 20% RH (use a filament dryer at 60 °C, or keep sealed in a container with desiccant for moisture protection)
Storage Conditions	≤ 30 °C, ≤ 20% RH (keep sealed with desiccant for moisture protection)
Compatible Support Material	Same material or dedicated PETG support material
Compatible Printer Type	Open-frame or enclosed
Compatible Nozzle Material	Brass, stainless steel, hardened steel, and other common nozzle materials
Compatible Nozzle Diameter	0.2, 0.4, 0.6, 0.8 mm
Compatible Build Plate	Textured PEI build plate, etc.
Build Plate Surface Preparation	Applying glue or adhesive spray is recommended
Recommended Nozzle Temperature	235-260 °C
Heated Bed Temperature	75-90 °C (depending on printer and build plate type)
Maximum Printing Speed	250 mm/s (255 °C, Bambu Lab X1C / P1S / A1)
Chamber Temperature	< 40 °C

• Properties

The commonly used physical, mechanical, chemical, and printing properties of TINMORRY Metallic PETG have been tested and are listed in the tables below.

• Physical Properties

Item	Test Method	Data
Density	ρ = m / v	1.30 g/cm³
Saturated Moisture Absorption	25 °C, 55% RH, ambient air	0.36%
Melt Flow Index	250 °C, 2.16 kg	12.7 ± 2.4 g/10 min

Glass Transition Temperature	DSC, 10 °C/min	71 °C
Crystallization Temperature	DSC, 10 °C/min	N/A
Melting Temperature	DSC, 10 °C/min	216 °C
Vicat Softening Temperature	ISO 306, GB/T 1633	75 °C
Heat Deflection Temperature 1	ISO 75, 1.82 MPa	65 °C
Heat Deflection Temperature 2	ISO 75, 0.45 MPa	71 °C

• Mechanical Properties

Item	Test Method	Data
Tensile Strength XY	ISO 527, GB/T 1040	49 ± 3 MPa
Tensile Strength Z	ISO 527, GB/T 1040	41 ± 3 MPa
Young's Modulus XY	ISO 527, GB/T 1040	1726 ± 138 MPa
Young's Modulus Z	ISO 527, GB/T 1040	1631 ± 182 MPa
Elongation at Break XY	ISO 527, GB/T 1040	8.2% ± 1.6%
Elongation at Break Z	ISO 527, GB/T 1040	3.7% ± 1.7%
Flexural Strength XY	ISO 178, GB/T 9341	71 ± 3 MPa
Flexural Strength Z	ISO 178, GB/T 9341	64 ± 2 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	2200 ± 148 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2144 ± 170 MPa
Impact Strength XY	ISO 179, GB/T 1043	26.7 ± 3 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	6.5 ± 1.8 kJ/m²

• Other Physical and Chemical Properties

Item	Data
Main Composition	PETG, additives
Skin Irritation	None
Water Resistance	Resistant
Oil and Grease Resistance	Resistant to most common oils and greases
Organic Solvent Resistance	Not resistant to certain organic solvents
Weak Acid Resistance	Resistant
Strong Acid Resistance	Not resistant
Weak Alkali Resistance	Resistant
Strong Alkali Resistance	Not resistant
Flammability	Flammable
Odor During Printing	Almost odorless
Odor During Combustion	Irritating odor

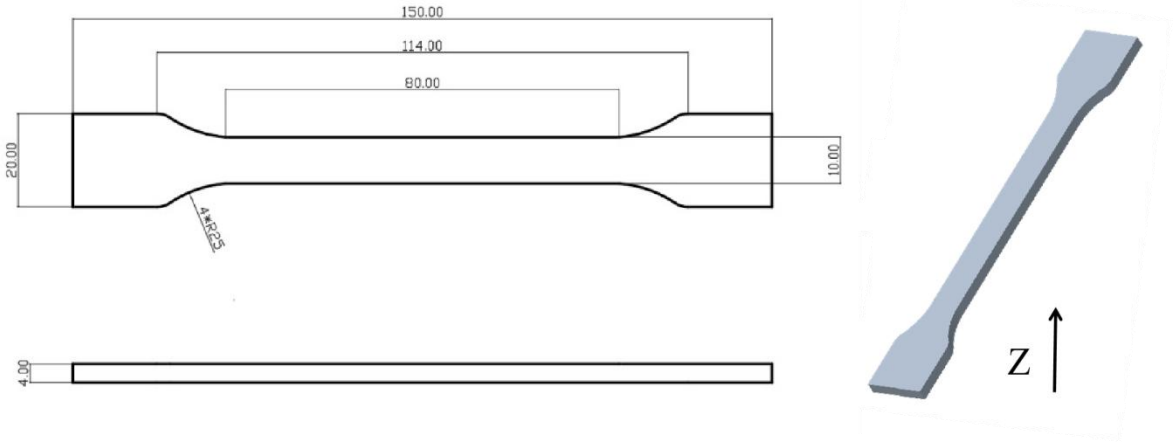
• Test Specimen Information

Mechanical Test Specimen Information	
Item	Data
Specimen Dimensions	See figure below
Nozzle Temperature	245 °C
Build Plate Temperature	80 °C
Printing Speed	150 mm/s
Infill Density	100%
Infill Pattern	Concentric

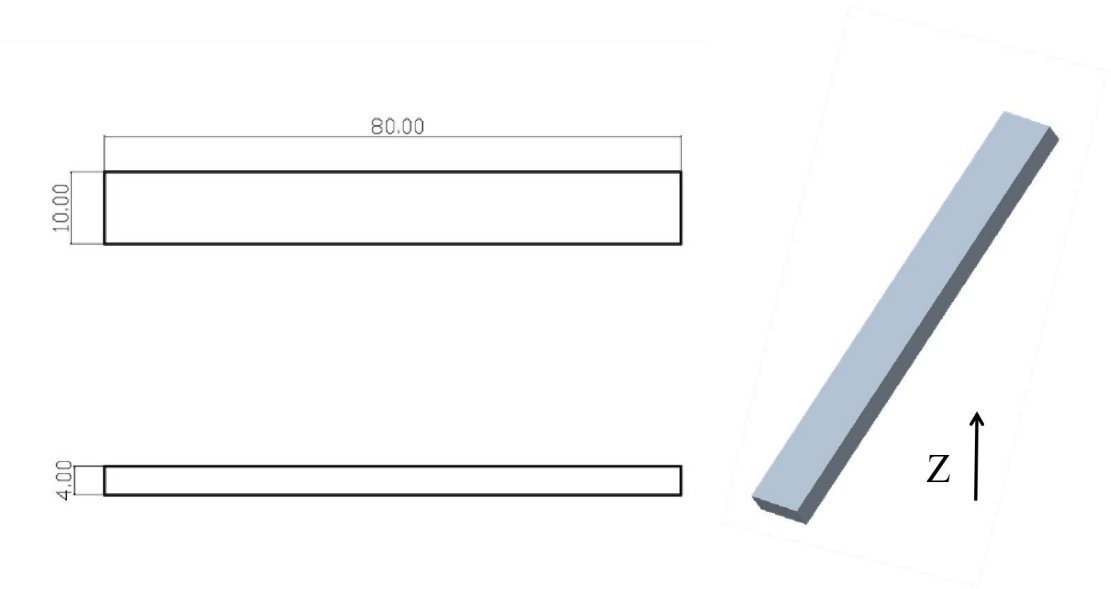
Notes:

1. The test specimens were printed using the key parameters listed above. Before testing, all specimens were annealed at 70 °C for 6 h and then conditioned indoors at approximately 25 °C for more than 12 h. Different ambient temperatures, fan speeds, and layer times during printing produce different cooling rates, which can affect the mechanical properties of the specimens, especially those printed in the Z direction. Different printers may also result in differences in printed-part properties.
2. For applications requiring higher print quality, thoroughly dry TINMORRY filament before printing and protect it from moisture during printing using a sealed container with desiccant (recommended relative humidity inside the container ≤ 25%) to reduce moisture-related printing issues. Refer to the "Recommended Printing Parameters" table for the recommended drying conditions. During drying, keep the filament and spool away from direct heat sources to avoid overheating damage. Do not use a kitchen oven or microwave oven.
3. Printed parts made from TINMORRY Metallic PETG show only limited improvement in strength, stiffness, heat resistance, and toughness after annealing, and some parts may shrink or deform during the process. The annealing result is also affected by part size, geometry, wall thickness, infill density, and other factors. If annealing is required, use a forced-air oven with uniform temperature. It is recommended to anneal at approximately 60 °C for about 2 h, then slowly increase the temperature to approximately 70 °C and hold for about 2-4 h. If obvious deformation occurs during the process, stop annealing immediately and use a lower annealing temperature for subsequent attempts.

1. Tensile Test



2. Flexural Test and Unnotched Impact Test



• Disclaimer

The data above were obtained by TINMORRY using specific printers, test methods, industry-standard specimens and methods, and specific equipment, and are provided for reference and comparison only. If the data change, TINMORRY will update this TDS, but may not be able to provide individual notice to users. In addition, the performance of 3D-printed parts depends not only on the intrinsic properties of the filament, but also on many other factors, including filament moisture content, printer type and condition, ambient temperature and ventilation, part geometry, and printing parameters. Users are responsible for the safety, legality, regulatory compliance, and performance of parts produced using TINMORRY 3D printing filament. TINMORRY is responsible for the quality of the filament sold, but assumes no responsibility for any damage or injury arising during its use or for the suitability of the filament for any particular application.